How-To Library

Informatica

Configuring Enterprise Data Preparation 10.2.2 and 10.4.0 on Amazon Elastic MapReduce with Autoscaling

© Copyright Informatica LLC 2019, 2022. Informatica, the Informatica logo, and Big Data Management are trademarks or registered trademarks of Informatica LLC in the United States and many jurisdictions throughout the world. A current list of Informatica trademarks is available on the web at https://www.informatica.com/trademarks.html

Abstract

You can configure Informatica Enterprise Data Preparation to run in the cloud on Amazon Elastic MapReduce (EMR). This article provides instructions and guidelines for configuring Enterprise Data Preparation to run on Amazon EMR. The article also describes how you can take advantage of autoscaling in an Amazon EMR cluster to improve Enterprise Data Preparation performance.

Supported Versions

- Enterprise Data Preparation 10.2.2
- Enterprise Data Preparation 10.4.0

Table of Contents

Overview. 2
Enterprise Data Preparation on Amazon EMR Architecture. 3
Prerequisites. 4
Prepare the Data Preparation Repository Database. 4
MySQL and MariaDB Database Requirements. 4
Oracle Database Requirements. 5
Configure Data Lake Access in Amazon Web Services. 5
Install Enterprise Data Preparation. 6
Create Hadoop File System Connections to the Data Lake. 6
Create Catalog Resources. 7
Create Resources for Amazon S3 Buckets. 7
Create a Hive Resource for Hive on S3. 7
Verify Enterprise Data Preparation on Amazon EMR. 8
Leveraging Amazon EMR Autoscaling for Enterprise Data Preparation. 8

Overview

You can configure Enterprise Data Preparation to run on Amazon Elastic MapReduce (Amazon EMR).

You should have knowledge of various Amazon Web Services components such as Amazon EMR, Amazon Relational Database Service (Amazon RDS), and Amazon Elastic Compute Cloud (Amazon EC2). You should also have knowledge of deployment practices for Informatica Big Data Management®, Enterprise Data Catalog, and Enterprise Data Preparation.

For more information about Amazon Web Services, see the Amazon product documentation.

To configure Enterprise Data Preparation on Amazon EMR, perform the following steps:

1. Complete the prerequisites.
2. Set up the Data Preparation Service repository database in Amazon RDS.
3. Configure data lake access in Amazon Web Services.
4. Install Enterprise Data Preparation in Amazon EC2.

5. Use Informatica Administrator to create a Hadoop File System connection to each Amazon S3 bucket in the data lake.
6. Use the Catalog Administrator to create Amazon S3, HDFS, and Hive resources for the data lake in Enterprise Data Catalog.
   You must create a resource for each Amazon S3 bucket in the data lake.
7. Verify that you correctly configured Enterprise Data Preparation for Amazon EMR.

For more information about installing and configuring Informatica Big Data products, see the Informatica® Big Data Suite 10.2.2 Installation and Configuration Guide.

You can take advantage of autoscaling in an Amazon EMR cluster to improve Enterprise Data Preparation performance. Autoscaling enables you to define threshold-based rules for adding and removing cluster task and core nodes. To leverage autoscaling, you must configure the Data Integration Service associated with Enterprise Data Preparation to run jobs on the Spark engine.

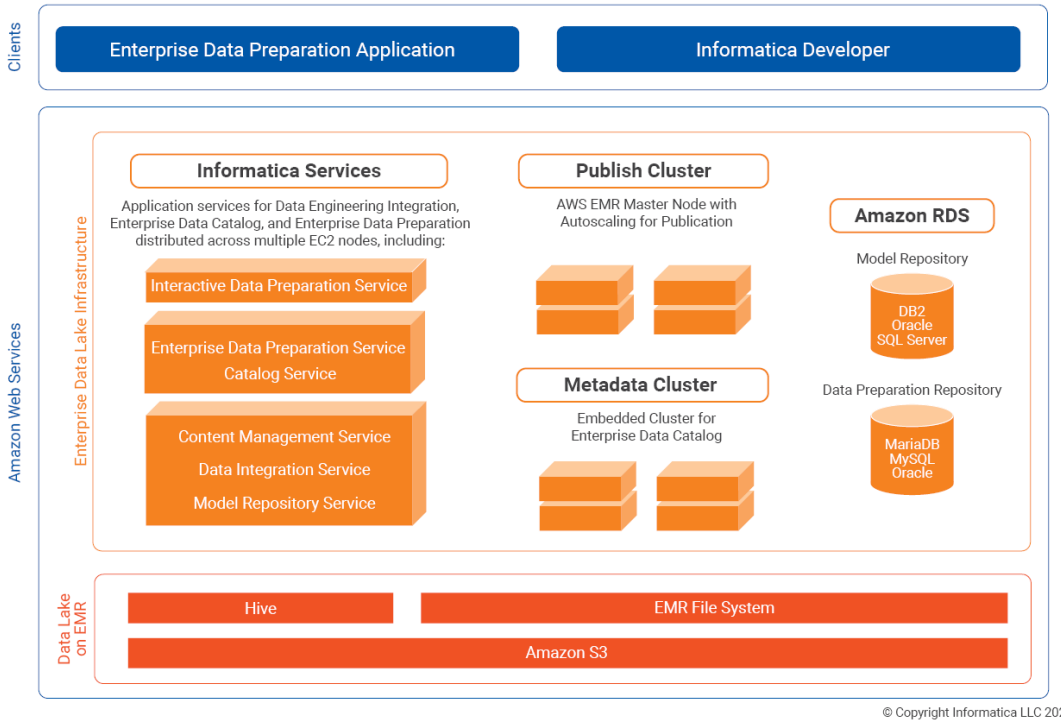
Informatica and the Amazon EMR cluster can use Kerberos authentication to verify user accounts when the Amazon EMR cluster supports Kerberos. You can use Kerberos authentication with the Informatica domain, with the Amazon EMR cluster, or with both.

Enterprise Data Preparation on Amazon EMR Architecture

When you deploy Enterprise Data Preparation on Amazon EMR, you install the Informatica domain, Enterprise Data Catalog, and Enterprise Data Preparation binaries on nodes running in Amazon EC2.

Enterprise Data Preparation on EMR uses your data lake as deployed on Amazon Web Services. The application goes through the EMR File System (EMRFS) to access data assets stored in one or more Amazon S3 buckets. The application uses Hive on S3 to enable querying and analysis of assets available in the data lake.

The following figure shows the deployment architecture for configuring Enterprise Data Preparation on Amazon EMR:



To leverage autoscaling, the HDFS, YARN, and Hive services must be available on an EMR cluster master node at all times.

Prerequisites

Complete the following prerequisites before you deploy Enterprise Data Preparation on Amazon EMR.

Enable Spark on the EMR cluster.

The Data Integration Service associated with the Enterprise Data Preparation Service runs jobs on the Spark execution engine on EMR nodes. When you install the Spark package, you can set up an external shuffle service on each worker node in the cluster to dynamically adjust the resources Enterprise Data Preparation uses.

Install the Informatica services.

Install and configure the Informatica services on an embedded Hadoop cluster running in Amazon EC2.

For information on installing the Informatica services, see the Informatica® Big Data Suite 10.2.2 Installation and Configuration Guide.

Complete the Hadoop integration tasks for EMR.

Complete the Hadoop integration steps described in the "Amazon EMR Integration Tasks" section in the Informatica 10.2.2 Big Data Management Integration Guide.

Install Enterprise Data Catalog.

Install and configure Enterprise Data Catalog on the embedded Hadoop cluster running in Amazon EC2.

For information about installing Enterprise Data Catalog, see the *Informatica 10.2.2 Enterprise Data Catalog Installation and Configuration Guide*.

Prepare the Data Preparation Repository Database

Set up the Data Preparation repository database in Amazon Relational Database Service (RDS). The Data Preparation repository stores worksheet metadata prepared by Enterprise Data Preparation users.

You can use a MySQL database, a MariaDB database, or an Oracle database as the Data Preparation repository database.

Allow 5 GB of disk space for the repository database. Allocate more space based on the amount of metadata you want to store.

MySQL and MariaDB Database Requirements

Perform the following steps to use a MySQL database or a MariaDB database as the Data Preparation repository.

Set the following system variables on the database server:

- For MySQL version 5.6.26 and higher, set lower_case_table_names=1.
- For MySQL version 5.7 and higher, set explicit_defaults_for_timestamp=1.

Ensure that the MySQL or MariaDB database has the following permissions:

- Create tables and views.
- Drop tables and views.
- Insert, update, and delete data.

The MySQL connector .jar file is not included with the installer. You must download the file and copy it to the following directory before you start the installer:

`$USER_INSTALL_DIR$/services/shared/jars/thirdparty/`

Make sure the name of the file is in the following format:

`mysql-connector-java-<versiondetails>.jar`

You must also set the mysql_connector_jar_path environment variable to the location of the MySQL connector .jar file when you run the installer to install Enterprise Data Preparation.

Oracle Database Requirements

Perform the following steps to use an Oracle database as the Data Preparation repository.

Ensure that the Oracle database has the following permissions:

- Create tables and views.
- Create sequence, session, and synonyms.
- Drop tables and views.
- Insert, update, and delete data.

Configure Data Lake Access in Amazon Web Services

Before you install Enterprise Data Preparation, configure AWS Identity and Access Management (IAM) roles to manage access to the data lake through the EMR File System (EMRFS).

Enterprise Data Preparation goes through EMRFS to access data assets stored in S3. EMRFS is an implementation of HDFS that all Amazon EMR clusters use for reading and writing regular files from Amazon EMR directly to Amazon S3.

If you use Amazon EMR release version 5.10.0 or later, you can use different IAM roles for EMRFS requests to Amazon S3 based on cluster users, groups, or the location of EMRFS data in Amazon S3.

Enterprise Data Preparation supports IAM roles to enable users to access assets stored in multiple Amazon S3 buckets. The default is EMR_EC2_DefaultRole, which you assign to nodes in the Informatica domain. You can configure a single IAM role that all users use to access the EMR cluster.

The following table describes the default IAM roles to select when you create EC2 and EMR instances:

Component	Default IAM Role
Informatica domain nodes	EMR_EC2_DefaultRole
EC2 instance profile for EMR cluster nodes	EC2 instance profile role
EMR cluster nodes	EMR_DefaultRole
Auto-scaling	EMR_AutoScaling_DefaultRole

Enterprise Data Preparation also uses EMRFS to access Hive on Amazon S3. You can configure roles to manage Hive access by different users.

Amazon EMR release version 5.10.0 and later supports the Kerberos authentication protocol. If the EMR cluster uses Kerberos authentication, you can enable fine-grained EMRFS authorization for Amazon S3 access on the cluster. You can create IAM role mappings for Kerberos users, Kerberos groups, or EMRFS data locations that provide access to Amazon S3 through EMRFS.

For more information about Amazon EMR security, see the Amazon EMR Management Guide.

Install Enterprise Data Preparation

Install Enterprise Data Preparation on the embedded Hadoop cluster running in Amazon EC2.

You can create the Enterprise Data Preparation Service and the Interactive Data Preparation Service during the installation process. If you want the installer to create the services, it creates both services on a node.

The installer prompts for cluster configuration associated with the Hadoop environment. The installer also prompts for the Data Preparation repository database connection details.

Informatica recommends that you associate dedicated Model Repository Service and Data Integration Service instances with the Enterprise Data Preparation Service. You can create a Model Repository Service and a Data Integration Service to associate with the Enterprise Data Preparation Service during installation.

When you install Enterprise Data Preparation on multiple nodes, you must install the Enterprise Data Preparation binaries in the Enterprise Data Catalog installation directory on a gateway node before you install the binaries on the other nodes. You must install the Enterprise Data Preparation binaries in the Enterprise Data Catalog installation directory to register the Enterprise Data Preparation Service and Interactive Data Preparation Service components with the Service Manager for the domain.

For information on installing Enterprise Data Preparation, see "Install Enterprise Data Preparation" in the Informatica Big Data Suite 10.2.2 Installation and Configuration Guide.

Create Hadoop File System Connections to the Data Lake

Use the Administrator tool to create a Hadoop File System connection to each Amazon S3 bucket in the data lake.

1. Click the **Connections** tab.
2. Select the domain in the Domain Navigator.
3. Select **Actions** > **New** > **Connection**.
4. Select **Hadoop File System**.
5. Configure the connection properties.

   The following table describes the connection properties:

Property	Description
Name	Name of the connection. The name is not case sensitive and must be unique within the domain. The name cannot exceed 128 characters, contain spaces, or contain the following special characters: ~ ` ! $ % ^ & * ( ) - + = { [}] | \ : ; " ' < , > . ? /
ID	String that the Data Integration Service uses to identify the connection. The ID is not case sensitive. It must be 255 characters or less and must be unique in the domain. You cannot change this property after you create the connection. Default value is the connection name.
Description	The description of the connection. The description cannot exceed 765 characters.
User Name	User name to access the storage system.

Property	Description
NameNode URI	The URI to access the S3 bucket. Format the URI as follows: `s3://<bucket name>`
Cluster Configuration	The name of the cluster configuration associated with the Hadoop cluster in EMR.

6. Click **Finish**.

Create Catalog Resources

Use the Catalog Administrator to create Amazon S3 and Hive resources for the data lake in Enterprise Data Catalog. You then use Enterprise Data Preparation to add the resources to the data lake.

A resource is a repository object that represents an external data source or metadata repository. Scanners attached to a resource extract metadata from the resource and store the metadata in Enterprise Data Catalog.

Create an Amazon S3 resource for each Amazon S3 bucket in the data lake. You must also create a Hive resource. The Hive resource is required to perform export, import, upload, data preparation, and asset publication operations in Enterprise Data Preparation.

For more information about creating resources and scanners, see "Creating a Resource" in the *Informatica 10.2.2 Catalog Administrator Guide*.

After you create the scanners, use the Enterprise Data Preparation user interface to add the scanners as resources to the data lake. You must add at least one Hive resource to the data lake.

For information about adding resources to the data lake, see the "Managing the Data Lake" chapter in the Informatica 10.2.2 Enterprise Data Preparation Administrator Guide.

Create Resources for Amazon S3 Buckets

Use Informatica Catalog Administrator to create an Amazon S3 resource for each Amazon S3 bucket in the data lake.

You can enter a virtual-hosted-style URL or a path-style URL as the value for the Amazon Web Services Bucket URL property. For example:

- virtual-hosted-style: `edlbucket.s3.amazonaws.com`
- path-style: `s3.amazonaws.com/edlbucket`

Create schedules for the resources you create so that Enterprise Data Catalog regularly scans the resources. As a best practice, schedule the resource scans to run during non-business hours.

For more information about creating an Amazon S3 resource, see the "Amazon S3 Resource Properties" section in the *Informatica 10.2.2 Catalog Administrator Guide*.

Create a Hive Resource for Hive on S3

Use Informatica Catalog Administrator to create a Hive resource that Enterprise Data Catalog uses to extract metadata from Hive tables in the data lake.

The Hive resource is required to perform export, import, upload, data preparation, and publish operations in Enterprise Data Preparation.

Observe the following guidelines when you create the Hive resource:

- In the **URL** property on the **General** > **Connection Properties** panel, specify the Fully Qualified Domain Name (FQDN) of the Hive server in the JDBC connection URL.
- If you are using operating system profiles, the Hive user name that you specify as the value for the **User** property must be a Hive superuser. For more information about operating system profiles, see the "Operating System Profiles" section in the Informatica 10.2.2 Enterprise Data Preparation Administrator Guide.
- Import the relevant connectors to extract metadata from Hive sources.

For more information about creating a Hive resource, see the "Hive Resource Prerequisites and Connection Properties" section in the *Informatica 10.2.2 Catalog Administrator Guide*.

Verify Enterprise Data Preparation on Amazon EMR

Verify that you can create and access Hive tables in Amazon EMR, create a Hive scanner to scan the Hive tables, and upload data assets to the Amazon EMR data lake.

1. Create a Hive scanner in Enterprise Data Catalog, and then configure the scanner to connect to the Amazon EMR cluster.

   For more information about creating resources in Enterprise Data Catalog, see the *Informatica 10.2 Enterprise Data Catalog Installation and Configuration Guide*.
2. Verify that you can access the data stored on Amazon S3 using Enterprise Data Preparation through Amazon EMR.
   a. Log in to Amazon EMR.
   b. Create a Hive table pointing to Amazon S3 using the following command:

      ```
      CREATE TABLE mydata (key STRING, value INT) ROW FORMAT DELIMITED FIELDS TERMINATED
      BY '=' LOCATION '{S3 location}';
      ```
3. Verify that the Hive scanner you created in the Enterprise Data Catalog finds the Hive table you created.
   a. Run the Hive scanner.
   b. Log in to Enterprise Data Preparation.
   c. Search for the Hive table you created.
4. Verify that you can upload a data asset to the data lake using Enterprise Data Preparation.
   a. Upload a file in Enterprise Data Preparation.

      When you upload the file, Enterprise Data Preparation writes the data to a Hive table in the data lake, and uploads the asset data to Amazon S3.

Leveraging Amazon EMR Autoscaling for Enterprise Data Preparation

You can take advantage of autoscaling in an Amazon EMR cluster to improve Enterprise Data Preparation performance.

Autoscaling occurs in Enterprise Data Preparation for the following activities:

When performing random sampling and filtering on large data assets during data preparation.

During data preparation, Enterprise Data Preparation submits Hive QL queries to Hive on the EMR cluster. Hive then converts the queries into jobs.

For assets smaller than 1 GB, Hive converts queries into Map/Reduce jobs. For data assets larger than 1 GB, Hive converts queries into Spark jobs.

When executing activities that involve mappings.

The Data Integration Service associated with the Enterprise Data Preparation Service runs Enterprise Data Preparation jobs on the Spark execution engine on EMR cluster nodes.

Define a scaling policy in EMR that contains threshold-based rules for adding and removing cluster task and core nodes. Use the AWS Management Console, the AWS CLI, or the Amazon EMR API to define a scaling policy.

Informatica recommends that you configure only task nodes with auto-scaling in Amazon EMR. For more information on configuring autoscaling in EMR, see the Amazon documentation.

Authors

Prathap Bailore

Rajiv Kambli

Dan Hynes